

Forkable Sandboxes

A Runtime Layer for AI Software Factories

What counts as new evidence when an agent loop repeats work?

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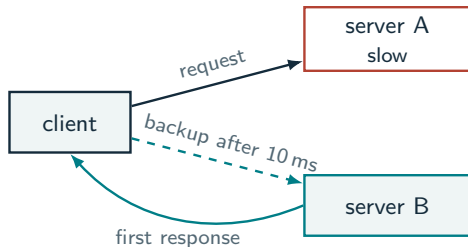
Disclosure: I work for a company that builds sandboxes.



A second request cut tail latency by $24\times$

- read 1,000 BigTable keys
- send a backup request after 10 ms
- accept the first response

99.9th-percentile latency	1,800 ms → 74 ms
additional requests	2%



The backup escapes a delay tied to the first server. Delays with a shared cause reach both servers and limit the benefit.

An additional execution helps when its cause of failure is separate.

Dean & Barroso, The Tail at Scale, CACM 56(2), 2013: backup requests after 10 ms, 99.9th-percentile latency for reading 1,000 BigTable keys, 2% more requests. The techniques assume the cause of variability does not hit several replicas at once.

More attempts found more solutions

SWE-bench Lite, issues solved by at least one attempt
(DeepSeek-Coder-V2-Instruct).

attempts per issue	issues solved
1 sample	15.9%
250 samples	56%
gain	40.1 points

- ✓ This measures coverage: some attempt in the collection is correct.
- ✗ It does not measure selection of that attempt by a production verifier.
- ✗ With an imperfect verifier and a cost for false acceptance, more sampling can reduce utility. The optimum is often fewer than ten attempts.

The remaining problem is to recognise the correct candidate.

Brown et al., Large Language Monkeys, arXiv:2407.21787, abstract. Stroebel, Kapoor & Narayanan, The Limits of Inference Scaling Through Resampling, arXiv:2411.17501, abstract: when false positives carry negative utility, the optimal number of attempts is often fewer than 10.

One question across six fields

What counts as a new, independent observation?



Physics and biology give vocabulary here and no evidence about forks.

In each field, apparently independent evidence is not independent.

Genomics and Bitcoin are team results with co-authors. Physics: PRE 2020, PNAS 2020, JPCB 2021. Genomics: Hitz et al. 2023, iScience 2026. Bitcoin: arXiv:2206.02871. Self-driving: perception work at Mobileye, 2021–2024. Agent loops: our experiments, arXiv:2607.09689. RL: Kimi K3 report §5.3.2.

Intelligence as choice and action

Intelligence derives from Latin *intelligere*: *inter* (between) and *legere* (to choose, to read).

Intelligence means choosing between.

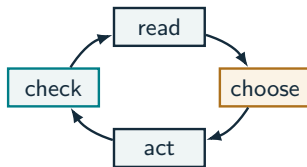
Since 29 Sep 2026, US federal agencies must say 'Super Intelligence'. We say intelligence.

ANALOGY This ladder is our own framing.

self-driving cars sense, plan and act

self-driving computers write and test their own software

self-driving intelligence is an agent in a loop



Choosing between forks requires knowing what each fork established.

Oxford English Dictionary / Online Etymology Dictionary, 'intelligence'. Executive order 'Inaugurating the Era of Super Intelligence', 29 Sep 2026 (White House fact sheet). Mobileye perception work, 2021–2024.

What this talk establishes

Question. What counts as new evidence when an agent loop repeats work?

Status	Evidence
MEASURED Observed	two work orders of our loop and their execution records
BUILT Implemented	evidence-ID rejection and lineage transport
PROPOSED Proposed	fork-cost and shared-ancestry experiments

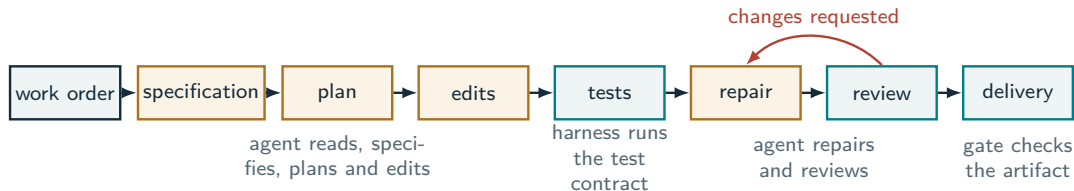
No run in this case study forked.

The incidents identify mechanisms worth fixing. They do not estimate failure frequencies, fork speedup or fork-induced correlation.

Outline. 2 Case study · 3 Design · 4 Analysis · 5 Evaluation · 6 Conclusion

Run records of two work orders, 27 Sep 2026. Reducer and evidence-ID check: Eliaz, arXiv:2607.09689.

The loop's execution path

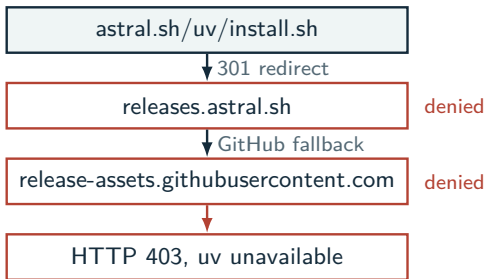


Sandbox	Docker Sandboxes, through Airflow's sandbox toolset
Model	one hosted model, served through Databricks, in every agent role
Approval gates	answered by the harness, recorded as "mode": "human"
Leases	not exercised (the cell was unmanaged)
Fork	unavailable: the work graph recorded <code>serial_fallback_missing_fork</code>

Each stage is a separate source of claims.

Our first work order, 27 Sep 2026: stage, approval and work-graph records. github.com/zozo123/ariflow-swfactory.

The installer needed two hosts the policy denied



initial allowlist	six hosts
missing destinations	two hosts
observed failure	HTTP 403

- The fix adds both hosts to code, scripts and documentation.
- A shared release-asset host admits more than this installer.

PROPOSED A proxy that fetches files by digest is a refinement. This run did not build it.

A hostname permission is an imperfect boundary.

Observed on our sandbox platform, 27 Sep 2026. It became our first work order (work-order record).

The approved plan named six files

The plan fixed the scope before any edit. Every planned file concerns the allowlist.

Planned change	Files
allowlist implementation: the Python check and the shell bootstrap	2
regression tests	1
deployment script	1
documentation	2
total	6

Acceptance conditions

- ✓ both redirect hosts present in every statement of the allowlist
- ✓ the policy statements agree with each other
- ✓ the regression checks pass

Not in the plan

- ✗ dispatch ownership logic

The plan defines what “outside the plan” means.

Our first work order, 27 Sep 2026: plan record and plan approval (answered by the harness).

A two-host fix became a 43-minute loop

Recorded stage	Work	Time	Cost
intent, specification, plan	acceptance conditions and the six-file plan	3:41	\$0.96
build and test	three edit tasks, suite failure, first repair	20:01	\$4.50
review	review, second repair, tests, re-review	18:55	\$4.87
recorded stages		42:38	\$10.33
timestamps, start to finish	11:39:56 to 12:23:40	43:44	

Two repairs and two reviews: 25.2 minutes and \$7.75, that is 59% of recorded stage time and 75% of reported model cost.

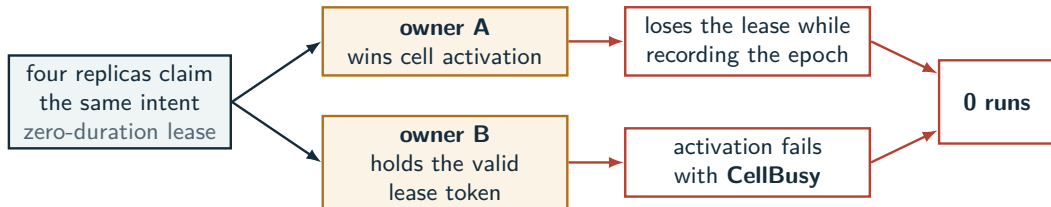
Times are agent-session durations, including tool activity. Sandbox setup took 20.3 s. A hundredfold faster setup would save about 20 s.

A failed test redirected the work into another subsystem.

Our first work order, 27 Sep 2026: stage metrics (2,557.5 s) and agent records, cost at list price.

Four replicas produced zero runs

Observed failure of a concurrency test: “0 runs from 4 replicas”.

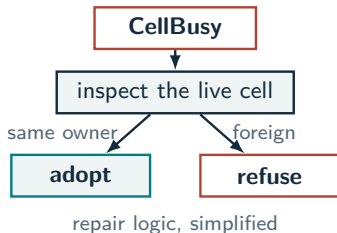


This is the first repair’s recorded diagnosis: a split-ownership race. One attempt owned the cell and another owned the dispatch, so neither dispatched a run. The incident alone does not validate the diagnosis.

A two-host allowlist fix led the repair into dispatch ownership.

Our first work order, 27 Sep 2026: build-stage test failure and the first repair’s agent record.

The new behaviour lacked a direct deterministic test



Step	Decision
review 1	blocker : add deterministic tests
repair 2	reuse a helper that other paths exercise
review 2	approve , and the missing branch test becomes <i>minor</i>

Reported suite: 1,972 tests including skips, zero failures. The harness ran it. The repair agent had no shell.

- Existing tests exercised adoption through a crash path and refusal through a foreign-owner busy path.
- No test deterministically established CellBusy together with same-owner adoption.

Both branches were tested separately. Their conjunction was not.

Our first work order, 27 Sep 2026: agent records of review 1, repair 2 and review 2; stage test counts.

Protected tests did not define permitted repairs

Rule	Consequence in this run
the repair stage protects tests	the agent reports it cannot add the requested test
unprotected source stays writable	the agent changes dispatch behaviour
planned steps receive scope checks	repairs follow a different path, without one
an out-of-plan edit is rated major	approval can still proceed

All 15 recorded edits were allowed, 6 of them by repairs. The test hook never fired.

PROPOSED Require a plan amendment when satisfying a review needs new behaviour or new tests.

ANALOGY OffRisk, built with our co-authors, **annotates** CRISPR off-target sites and their risk. A scope check would **enforce** a permitted scope.

Test protection and repair scope were separate controls.

First work order, 27 Sep 2026: edit-hook log, agent records. Barkai, Malul, Eliaz et al., OffRisk, *Bioinf. Adv.* 3:vbad138 (2023).

Visible feedback can become an optimisation target

Read-only tests prevent test edits. They do not establish behavioural coverage.

Source	Reported finding
METR (2025)	reward hacking in 30.4% of RE-Bench runs and 0.7% of HCAST runs. Visibility of the scorer is a hypothesis
ImpossibleBench	read-only tests do not prevent special-casing of the code for the tests
Kimi K3 §4.2.6	public diagnostic verifiers are paired with hidden held-out verifiers

Public diagnostic feedback supports repair. Reserved validation tests the result. Our incident shows an unresolved test obligation. It shows no deliberate cheating.

Diagnostic feedback and reserved validation serve different purposes.

METR (June 2025); ImpossibleBench (arXiv:2510.20270); Kimi K3 technical report §4.2.6.

The gate refused correctly, and cleanup lost the result



Our second work order: about 26 minutes, \$3.43, no delivered pull request. The unexpected file was our harness's own review-diff archive, mirrored into the workspace.

PROPOSED Before teardown, export the patch, input identity, evidence and delivery state. During a long human wait, preserve or snapshot the state without granting publication permission.

A correct refusal and a failed recovery are separate events.

Our second work order, Sep 2026: run records (about 26 min wall clock, 853.8 s stage time, \$3.43); redacted copy on request; fix proposed, not merged. Related: Planarian statepoints (arXiv:2609.35366); Crab turn-level recovery (arXiv:2604.28138); DeepSeek-V4 DSec (arXiv:2606.19348 §5.2.5).

Three receipts misrepresented their events

Recorded claim	What happened
CI success	evaluation steps were skipped because a key was absent, and the job passed in 3s
Human approval	the harness answered the gate, and the record says "mode": "human", actor admin
Structured CLI result	status text contaminated stdout, and an earlier attempt failed while parsing it

Required states

executed skipped failed

Required actor

human harness service

A trusted writer can still create a semantically incorrect receipt.

A receipt must state what ran and who acted.

Our CI, 28 Sep 2026; our work orders, Sep 2026: approval records and setup logs. Provenance: PASS (USENIX ATC'06); CamFlow (SoCC'17); in-toto (USENIX Security'19).

We recorded the adapter digest, but not its recoverable source

Recorded in the run

- ✓ input commit and dependency lock
- ✓ policy digest
- ✓ every agent step's output
- ✓ sandbox adapter digest

Missing

- ✗ adapter source in the repository: a local shim outside Git

Audit	inspect the recorded run
Replay	reproduce execution against recorded external responses
Fresh run	obtain new external responses, including new hosted-model calls

A digest identifies what ran. It does not reconstruct it.

Our first work order, Sep 2026: run record. ENCODE records reference, software and parameters per data file: [Hitz et al., bioRxiv 2023](#). Record and replay: rr (USENIX ATC'17); DeepSeek-V4 DSec ([arXiv:2606.19348 §5.2.5](#)).

A snapshot captures part of the experiment

State surface	Record needed
repository and dependencies	tree and dependency identities
guest files and processes	declared snapshot semantics
model continuation	prompts and responses
evaluation	test, fixture, seed and feedback exposure
external services	requests and results

Known mechanisms: OBuilder snapshots each build step. Overlay layers preserve filesystem state. Process and VM checkpoints preserve more.

Two children with identical files can receive different hosted-model responses. Different children can reuse the same evaluation feedback.

A successful restore does not establish complete experimental fidelity.

OBuilder; overlay layers: [day10](#), [DeltaBox \(arXiv:2605.22781\)](#); DeepSeek-V4 DSec ([arXiv:2606.19348 §5.2.5](#)).

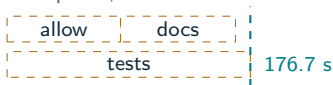
Parallel editing offered a computed 166-second saving

Task	Time	Depends on
allowlist	82.4 s	none
tests	176.7 s	none
documentation	83.5 s	allowlist
serial, as run	342.6 s	
critical path	176.7 s	computed
difference	165.9 s	6.5% of stage time

as run

allow	tests	docs
-------	-------	------

computed, no fork cost



reason: `serial_fallback_missing_fork` recorded by the executor. The computed schedule assumes unchanged durations and excludes fork, merge, contention and extra verification. Separate worktrees could support it.

A live fork must justify itself by saving expensive reached state.

Our first work order, Sep 2026: work-graph record (durations measured, schedule computed); stages 2,557.5 s.

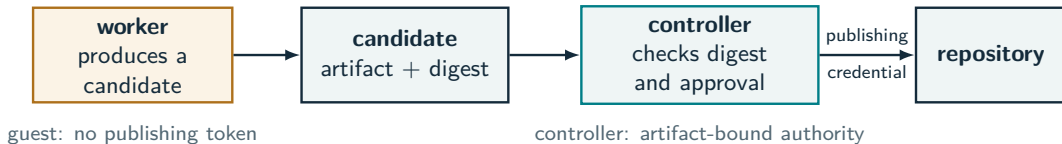
Seven surfaces produced concrete runtime requirements

Surface	Requirement	Addressed in Section 3 by
networking and credentials	make permitted access explicit	credential kept by the controller
filesystem state	identify the candidate's input state	immutable, identified parent
build and tests	record the exact behavioural obligation	evaluator outside the worker
recovery	preserve artifacts before cleanup	export, then tear down
observability	distinguish execution from skipped work	artifact-bound evidence
reproducibility	preserve adapters and external responses	recorded inputs and observations
cloning	declare what children share	fork with declared sharing

Each incident identifies a mechanism. None estimates a failure rate.

Our two work orders, Sep 2026: run records. Mechanisms are proposed unless Section 3 marks them built or exercised.

Workers produce candidates and the controller authorises publication



✓ **BUILT** Built: the coding cell holds no publishing credential, and delivery follows the gate.

✗ A generation string supplied by the guest cannot authenticate a child: a fork copies it.

PROPOSED Restore rule: the host assigns each child an identity, a fresh generation, an expiry and a budget.

Publishing authority stays with the controller, bound to one artifact.

Our first work order, 27 Sep 2026: approval and delivery records; Gray & Cheriton, Leases, SOSP'89.

Kernel isolation and authority are separate controls

Configuration	Kernel separation	Publishing authority
microVM containing a bearer token worker, with controller-mediated publication	hypervisor boundary depends on the runtime	a copied token can remain usable the controller checks each publication

Kernel surface: containers, gVisor, microVMs and unikernels expose different amounts of kernel.

Explicit authority: Capsicum and CHERI give a process only the rights it is handed.

PROPOSED Sharing policy for children

- warm dependency cache: immutable, identified by digest, shared within one trust domain
- writable branch state: private to the child

Immutability does not remove timing or content leakage, so a shared cache still needs a trust policy.

Firecracker (NSDI'20); gVisor documentation; Watson et al., Capsicum (USENIX Sec'10) and CHERI (IEEE S&P'15).

VM fork already had an API in 2009

SnowFlock, parallel computation

```
ID = VM_fork(N)
if ID.isChild():
    parallel_work(data[ID])
    VM_exit()
else:
    VM_wait(ID)
```

Agent pattern, our reading

```
ID = fork(reached_state, N)
if ID.isChild():
    generate_continuation()
    exit()
else:
    evaluate(candidates)
```

Caller

Responsibility

data-sharded

assigns distinct work to each child

searching

interprets candidates that may share assumptions and checks

The mechanism is old. The responsibility of the caller has changed.

Lagar-Cavilla et al., SnowFlock: rapid virtual machine cloning for cloud computing, EuroSys'09, Fig. 1(b); built on Xen (Barham et al., SOSP'03).

Cloning, checkpointing and recovery evolved differently

Mechanism	Systems	Main concern
virtualisation and migration	Xen (SOSP'03), live migration (NSDI'05), Potemkin (SOSP'05)	isolation, moving and cloning VMs
state cloning	SnowFlock (EuroSys'09), Catalyzer (ASPLOS'20), Nephele (EuroSys'23), MITOSIS (OSDI'23)	fast start from existing state
agent checkpointing	DeltaBox, Shepherd (2026)	cheap rollback and branching of agent runs
recovery and replay	Crab, Planarian, DeepSeek DSec (2026)	failures and external effects

Recovery systems are not interchangeable fork implementations.

SOSP'03; NSDI'05; SOSP'05; EuroSys'09; ASPLOS'20; EuroSys'23; OSDI'23; [DeltaBox](#); [Shepherd](#); [Crab](#); [Planarian](#); [DeepSeek-V4](#), [arXiv:2606.19348](#) §5.2.5.

Published runtimes report different operations

System	Operation reported	Value	Source
DeltaBox	checkpoint	10.83 ms	arXiv 2026
Kimi K3 AgentENV	checkpoint	as low as 133 ms	tech. report §5.3.2
Kimi K3 AgentENV	resume	as low as 49 ms	tech. report §5.3.2
Shepherd	fork	134–143 ms	arXiv 2026
SnowFlock	remote fork across hosts	600–800 ms	EuroSys'09
Firecracker	boot to application code	under 125 ms	NSDI'20

Each value needs its operation, workload, endpoint and timing convention. The rows cannot rank platforms.

Survey gap: none of the systems examined measures whether a child's correctness depends on its siblings.

DeltaBox; Kimi K3 §5.3.2; Shepherd; Lagar-Cavilla et al., EuroSys'09; Agache et al., Firecracker, NSDI'20.

Training systems fork reached state for evaluation

51.2 million

sandboxes, across 1.51 million
images

Kimi K3, training and evaluation

- **pause** the sandbox while the model generates, up to 98% of its life
- **fork** the sandbox “for reward judging without side effects”
- **snapshot** at intervals, for error recovery
- **partial rollout**: update once a fraction λ of trajectories completes, to limit long-tail latency

DeepSeek-V4's DSec runs hundreds of thousands of concurrent sandboxes per cluster. It logs commands and replays cached results when a training task resumes.

The sandbox is forked for judging. The judge itself is not necessarily forked.

They show infrastructure use. They do not show independent rewards.

Kimi K3 §4.1.2, §5.3.2 (AgentENV); DeepSeek-AI, DeepSeek-V4, [arXiv:2606.19348](#) §5.2.5.

Checkpoint success is conditional on reaching the checkpoint

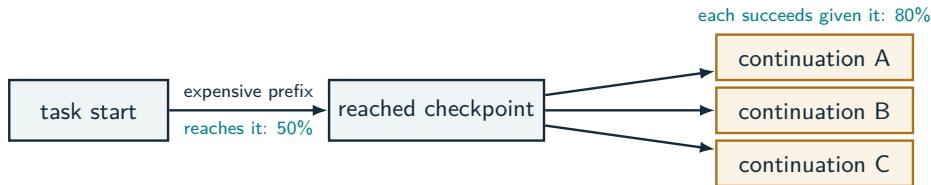


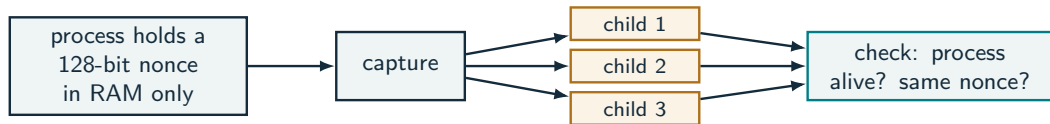
Illustration. end-to-end success = $0.5 \times 0.8 = 0.4$, while a checkpoint-started run shows 0.8.

- A checkpoint distribution differs from the task-start distribution.
- When every repair inherits one plan, successful continuations cannot validate that plan.
- Go-Explore makes reuse of reached states explicit: first return, then explore.

Successes after a checkpoint say nothing about the prefix they share.

Ecoffet et al., *First return, then explore*, Nature (2021). Probabilities are illustrative, not measured.

The snapshot probe defines the claimed state surface



Result

Permitted description

files survive only

filesystem restore

process and RAM survive

evidence of process and memory restoration

declared semantic checks match

fidelity for those checked properties only

Passing is necessary for a memory claim. Full fidelity needs more. Compare declared outputs, ignoring timestamps. Worktrees, CRIU, VM snapshots and replay logs restore different state. Kimi reports up to $6.5\times$ memory overcommit from copy-on-write plus page-cache optimisations.

Gate 0 is preregistered and unrun, so we say restore fan-out.

Our preregistered protocol, Gate 0. [Kimi K3 §5.3.2](#). CRIU and Firecracker snapshot documentation.

Children can inherit secrets, identities and random state

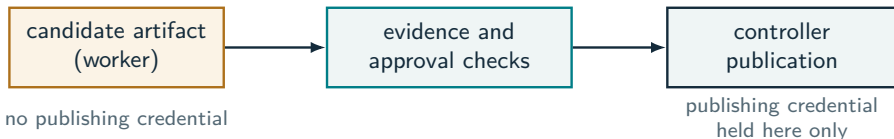
Inherited state	Required treatment	What is known
bearer tokens in memory random-number state	keep publishing secrets outside the guest declare copying or reseeding	every child holds a live copy VMGenID reseeds the kernel, userspace has no generic solution
guest clock IP and MAC identity open connections	resynchronise, or model explicitly assign child-specific networking declare the supported restore semantics	the child resumes at snapshot time clones otherwise share one identity CRIU restores a connection for one restore, gVisor can reset connections

Eight clones can hold eight copies of one live token.

Divergence costs memory: each page a child writes becomes its private copy.

Firecracker snapshot documentation (VMGenID, random state); [Brooker et al., arXiv:2102.12892](#); CRIU and gVisor checkpoint/restore documentation.

Publication has an authoritative commit path



- ✓ **MEASURED** exercised: the work cell held no publishing credential, and delivery followed the gate
- ✗ model requests leave the sandbox immediately and incur cost
- ✗ an already-sent tool request cannot be undone (Zheng et al. 2026)
- **PROPOSED** a general effect buffer, with reconciliation or idempotency for uncertain effects

Speculator, external synchrony and Remus delay output until it is safe. DSec replays cached results on resumption.

Nightingale et al. SOSP'05 (Speculator) and OSDI'06 (external synchrony); Cully et al. NSDI'08 (Remus); Zheng et al., [arXiv:2608.22928](#); DeepSeek-V4 [arXiv:2606.19348](#) §5.2.5.

Fork saves resources when preparation dominates reuse cost

$$(N - 1) P > H + N (R + D)$$

preparation P , capture H , restore R , divergence D , children N . Additive resource-cost model.

Calculation, not a measurement. Units: resource-seconds.

Case	Avoided preparation	Added reuse cost	Result
$N=8, P=30, H=2, R=1, D=3$	$7 \times 30 = 210$	$2 + 8 \times 4 = 34$	fork saves 176
same, warm cache $P=2$	$7 \times 2 = 14$	34	fork loses 20

Batch elapsed time is a separate question: it depends on critical paths and contention. A faster-booting minimal image is the alternative (Jitsu, LightVM).

Madhavapeddy et al. NSDI'15 (Jitsu); Manco et al. SOSP'17 (LightVM, NEC Labs). Parameter values are illustrative.

Our API measurements cover three different workflows

Workflow	Success	p50	p95
Daytona: create	1,024/1,024	0.20 s	1.12 s
Tensorlake: create	256/256	3.44 s	9.00 s
Our sandbox platform: restore, run, capture	255/256	6.87 s	9.04 s

- Restore path: a 141 MB named snapshot at concurrency 12
 - Whether the snapshot captures memory is unverified, so the third row is restore fan-out
 - Per-operation teardown is excluded from the percentiles
- ✗ different operations and workloads: API round trips, not a ranking and not a memory-fork latency

The table establishes exercised API paths and their latencies.

Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689), Table 2.

Separate calls represent separate decisions

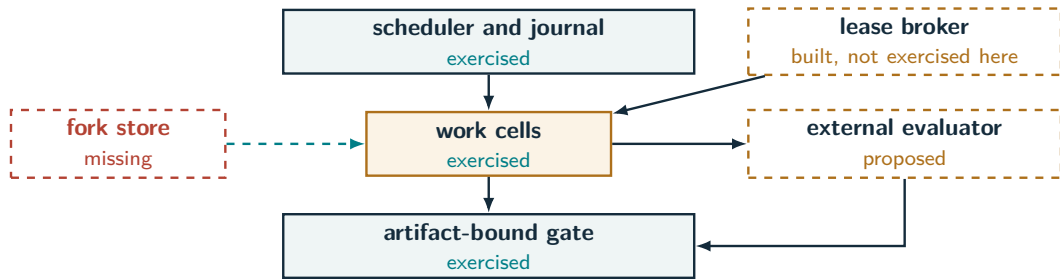
Call	Responsibility	Status
checkpoint	identify the captured state	MEASURED named snapshots exercised
fork	create a child with declared sharing	PROPOSED
evaluate	produce artifact-bound evidence	PROPOSED external evaluator
select	choose one candidate	PROPOSED
reduce	combine measurements	BUILT deduplication
promote	authorise publication	BUILT digest gate

- fork declares the seed policy, the lease identity and the generation
- ✗ the built reducer carries lineage, without dependence-adjusted uncertainty or abstention
- a fresh test is a separate evaluation event, outside select

Select chooses a candidate. Reduce combines measurements.

Reducer: Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689). Fork contract and interface: our design documents.

The loop has a gate, but no fork store



Exercised path: work order → scheduler → one work cell → gate → publication.

Proposed branch path: fork store → child cells with leases → external evaluator → gate.

The runtime contract was written before any fork implementation.

Our design documents: fork lifecycle and run recovery. Solid: exercised in our runs. Dashed gold: built but unexercised, or proposed. Dashed red: missing.

Each branch needs a complete lifecycle contract

Phase	Condition	Status
Creation	1 immutable, identified parent	proposed
	2 child identity, lease, expiry and budget	proposed (lease broker built)
Execution	3 no publishing credential in the guest	enforced in our runs
	4 record inputs, ancestry, observations and feedback	recorded in part (lineage)
Decision	5 reject reused observations, record shared factors	rejection enforced in the reducer
	6 compose deterministically, retain conflicts	proposed
	7 verify the exact selected composition	proposed
Recovery	8 export artifacts and evidence, then tear down	teardown observed, export proposed

Export before teardown keeps the result and grants no authority.

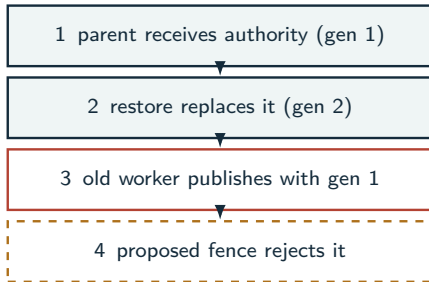
Enforced: a check that ran. Observed: seen in our runs. Recorded: carried with results. Proposed: designed, not run. Our fork-contract design document. Reducer: Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689).

Promotion invariants exist, and no model check has run

Invariants

- evidence refers to the frozen candidate
- approval refers to that same candidate
- current authority governs publication
- a stale generation cannot publish
- an uncertain effect cannot be blindly retried

Stale-generation sequence



A TLA+ design model exists. No TLC run and no implementation trace check is recorded.

Dean and Barroso note that consistent updates use quorum algorithms such as Paxos.

TLA+ design model of our loop's promotion (design document). Dean & Barroso, The tail at scale, CACM 2013.

Candidate selection and measurement pooling answer different questions

	Nine candidate patches	Nine measurements of one candidate
Object	nine different artifacts	one frozen artifact
Purpose	choose which patch to validate	estimate one defined quantity
Risk	selection can overfit the feedback	outcomes can be dependent
Requirement	verify the selected result on fresh cases	model the uncertainty, including dependence

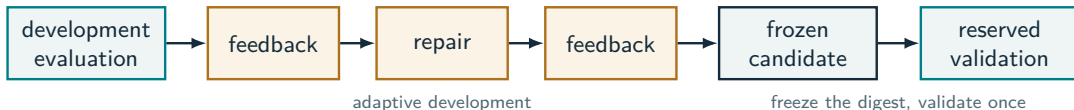
Different patches estimate different quantities. Their summaries cannot be pooled as one parameter.

In our run, one hosted model reviewed two successive repairs of one patch. The reviews guided the repair. They were not independent validation.

Repeated reviews can improve a patch without validating it independently.

Our experiments, Sep 2026: records of repair 1, review 1, repair 2 and review 2.

Development feedback changes the candidate being evaluated



OUR RUN

Review 1 requested changes with a blocker.
Repair 2 responded to it. Review 2 approved.
One hosted model performed all four steps.

PUBLISHED

Reused holdouts overfit (Dwork et al. 2015).
Adaptive analysis biases estimates (Russo & Zou 2016). Kimi K3 pairs public diagnostic verifiers with hidden held-out ones.

PROPOSED Record which feedback each candidate received. Bind final validation to the frozen digest.

Feedback used for repair cannot also serve as the final validation.

Our experiments, Sep 2026. Dwork et al., Science 2015. Russo & Zou, AISTATS 2016. Kimi K3 §4.2.6.

Four reruns do not create four times as many test cases

Event	What is new
resubmit an old receipt	no new observation
rerun 100 deterministic cases	new executions, same cases
rerun with fresh scheduler randomness	new outcomes within repeated-case clusters
evaluate new cases	new evaluation conditions

Illustration. Four runs of 100 cases give 400 outcomes in 100 case clusters.

Keep four identifiers separate: **execution ID**, **observation ID**, **test-case ID**, **candidate ID**.

ANALOGY · BIOLOGY Our 2021 CRISPR screen: ≥ 4 technical replicates per point, ≥ 2 independent experiments per plot.

Deduplicate reused observations. Model dependence among new ones.

Gilad, Eliaz et al., Commun. Biol. 4:399 (2021), Methods.

Pairwise correlation does not determine best-of- N coverage

Calculation. Three candidates. In each vector, 1 = candidate wrong. Listed vectors are equally likely.

Error vectors	Individual error rate	Pairwise correlation	All wrong
{000, 011, 101, 110}	50%	0	0%
all eight vectors	50%	0	12.5%
{001, 010, 100, 111}	50%	0	25%

Same marginals and zero pairwise correlation, yet the all-wrong rate ranges from 0% to 25%.

A selector that returns one candidate's answer succeeds with probability at most $1 - P(\text{all wrong})$.

An effective sample size describes the precision of an average. It is not a guarantee for search.

Proposed: record every candidate's pass or fail.

Calculation by enumeration. Ceiling: [Chen, arXiv:2606.27288 \(2026 preprint\)](#), ceiling $1 - \beta$, β the all-fail rate.

Correlation raises the variance of an average

Y_i : one declared outcome of one frozen candidate under a fixed evaluation protocol.

Equal marginal variance σ^2 , common pairwise correlation $\rho \geq 0$:

$$\text{Var}(\bar{Y}_N) = \sigma^2 \left(\rho + \frac{1 - \rho}{N} \right)$$

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\rho}$$

Variance-equivalent sample size measures precision. Shared bias remains.

ANALOGY Amdahl's speedup has the same form, $N/(1 + (N - 1)f)$.

Examples that assume independent draws

- Dean & Barroso: 1 slow request in 100 per server, 100-way fan-out: $1 - 0.99^{100} \approx 63\%$ of requests slow.
- Hypothetical race failing 1 run in 10: 29 green runs give $0.9^{29} = 0.047 < 0.05$.
- Dorfman 1943 pooling. Poolkeh: a model of 295,951 tests for ~ 9 M people.

State the population of draws before counting them as independent.

Kish (1965). Amdahl (1967). Dean & Barroso, CACM 2013. Dorfman (1943). Poolkeh, medRxiv 2020.

One hundred measurements can have the precision of nine

Calculation. $N_{\text{eff}} = N/(1 + (N - 1)\rho)$

ρ	$N = 9$	$N = 100$
0	9.0	100.0
0.05	6.4	16.8
0.10	5.0	9.2
0.50	1.8	2.0
1.00	1.0	1.0

SEPARATE EMPIRICAL COMPARISON

Nine LLM judges from seven model families: Kish $N_{\text{eff}} = 2.18$, 95% interval 2.07–2.31. A 2026 preprint on judge panels. It is not a fork measurement.

Kim et al. report another quantity: two models that both miss a question give the same wrong answer 60% of the time, against 33% by chance. That is a conditional agreement rate. It is not ρ .

At $\rho = 0.1$, one hundred measurements carry the precision of about nine.

Calculation from Kish (1965). Kohli, arXiv:2605.29800 (2026 preprint). Kim, Garg, Peng & Garg, ICML 2025, HELM, MMLU.

Runtime manifests identify shared factors

Manifest fields per result

candidate_digest	parent_snapshot
execution_id	input_tree
observation_ids	prompt_digest
model_version	seed_policy
test_and_fixture_ids	feedback_ids

BUILT Lineage travels with each result.

PROPOSED The full field set and its use in analysis.

Three kinds of sharing

- ✓ known duplication: a reused observation ID
- ✓ shared ancestry: one parent, prompt, test or seed
- ✗ unknown blind spots shared by models

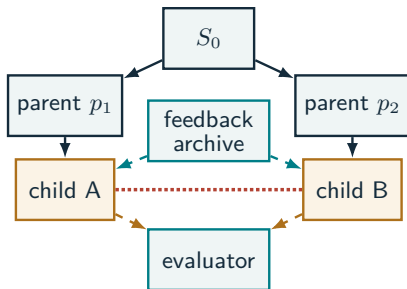
Recorded sharing is metadata. Its effect requires measurement.

Our co-authored studies: address linking tied most early bitcoin mining to 64 agents. Malignant-cell clustering depended on patient identity. Neither measures a fork correlation.

The unit of analysis and its cluster labels must be recorded with each result.

Green, Karvounarakis & Tannen, Provenance semirings, PODS'07. Blackburn et al., arXiv:2206.02871 (2022). Saurty-Seerunghen et al., iScience 2026. Eliaz, arXiv:2607.09689.

Ancestry is a tree and evidence flow is a graph



Edge labels

- > parent relation: state ancestry
- -> shared feedback: information influence
- -> shared evaluator: information influence
- composition: interaction of selected patches

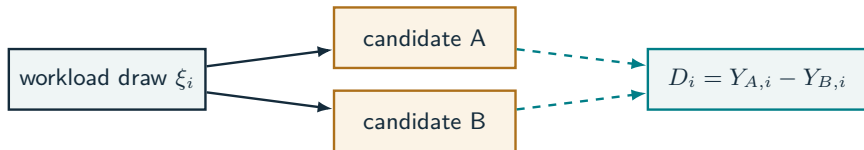
A and B have different parents but share two sources of information.

ANALOGY · PHYSICS Our work on linker structure, branching and motor-driven avalanches motivated attention to connectivity. Stockmayer's gelation theory (1943) holds under its own assumptions. It does not predict an agent threshold.

Eliasz et al., PRE 2020. Liman et al., PNAS 2020. Li et al., JPCB 2021. [Stockmayer, J. Chem. Phys. 11:45 \(1943\)](#), analogy only.

Shared conditions can improve paired comparisons

$$\text{Var}(Y_A - Y_B) = \text{Var}(Y_A) + \text{Var}(Y_B) - 2 \text{Cov}(Y_A, Y_B)$$



Repeat the pair across fresh workload draws $i = 1, \dots, n$.

Purpose	Design
Compare A with B	Match relevant conditions
Assess robustness	Vary relevant conditions

PROPOSED The fork API should state which randomness it copies, which it reseeds and which it leaves external.

Common random numbers: [Glasserman & Yao, Management Science 38\(6\) \(1992\)](#).

Passing pairs can conceal a failing composition

Illustration

Three patches each enable one more worker.

Capacity permits two workers.

Patches applied	Workers	Capacity 2
$\{1\}, \{2\}, \{3\}$	1	✓ pass
$\{1, 2\}, \{1, 3\}, \{2, 3\}$	2	✓ pass
$\{1, 2, 3\}$	3	✗ fail

$$\Gamma_{ij} = U(\{i, j\}) - U(\{i\}) - U(\{j\}) + U(\emptyset)$$

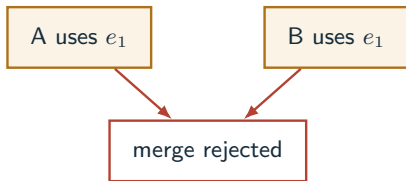
- $U(A)$: a fixed objective for patch set A on one parent.
- With $U = 1$ for pass and 0 for fail, every pair has $\Gamma_{ij} = 0$.
- Eight candidates give 28 pairs and 256 subsets.
- **PROPOSED** Rebuild and evaluate the exact composition selected for publication (contract condition 7).

ANALOGY · BIOLOGY Genetics calls a strongly negative interaction synthetic lethality. We screened for gene knockouts that are lethal in combination with an SRC-3 inhibitor.

Gilad, Eliaz et al., Commun. Biol. 4:399 (2021).

The built reducer rejects reused evidence IDs

Case 1: one observation, two workers

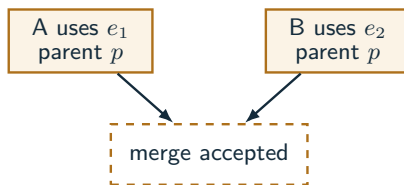


Duplicate evidence ID e_1 .

BUILT

- duplicate evidence IDs stop the merge
- lineage travels with the result

Case 2: two observations, one parent



Distinct observations. Dependence through p unresolved.

PROPOSED open

- dependence-adjusted uncertainty
- abstention when sharing is unknown

A new execution ID does not make copied data fresh.

Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), §2–5. Cochran, Biometrics (1954), the inverse-variance estimator.

Synthetic checks exercise numerical pooling

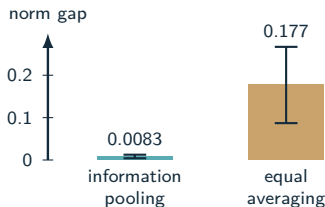
End-to-end trace, four workers

Four concurrent restore, run and capture round trips on our sandbox platform, timed by the client.

Pooled estimate	4.9422
Full-sample estimate	4.9450
Absolute difference	0.0028
End-to-end trace	6.70 s

Five logistic shards, $n = 60$ to 5,000

Distance to the full-data MLE, lower is better.



Assumes a common parameter and calibrated information. Mean $\pm$ SD over eight fixed seeds, not confidence intervals. Sample-size weighting was not compared.

These checks test aggregation, not agent-task accuracy or dependence correction.

Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), §4: synthetic checks and trace.

Claimed precision can be inflated

One shard inflates its precision

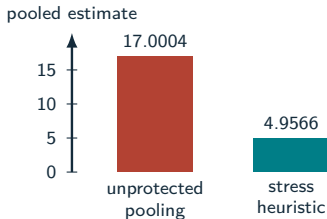
A 2,000-point shard returns a distant estimate and inflates its reported precision a further $50\times$.

- Claimed precision raises the shard's weight.
- Sample count alone cannot detect inflated information per point.
- The stress heuristic is not a Byzantine guarantee.

PROPOSED The controller sets each shard's precision by trusted recomputation or calibrated external scoring.

Precision must come from outside the worker.

Synthetic check, Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), §4. Not an agent-task benchmark.



Test 1 compares restore with a warm cached template

Pre-registered and unrun. No model calls.

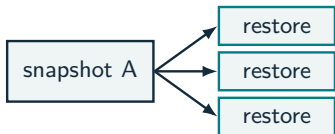
Arm	Starting point
Warm template	dependencies baked in, source fetched at the pinned commit
Restore	captured reached state after setup and one warm test run
Cold reference	no cache, used as a reference only
Checks first	record whether a RAM nonce survives (else report fan-out), and restored and direct outputs match
Design	$N \in \{3, 6, 12\}$ concurrent cells, 20 interleaved batches per arm and N
Outcome	time from the batch request until the last child's first test result
Supports	lower bound of the 95% CI of the median gain above 10s at every N
Rejects	upper bound of the 98.3% (Bonferroni) CI below 10s at any N

A result in which the cache wins is a valid outcome.

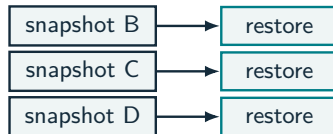
Pre-registration, tags prereg-v1 to prereg-v3. Restore failures are reported with the timing results.

Test 2 measures environmental co-failure

Pre-registered and unrun. No model calls.



shared parent
three restores from one snapshot



separate preparation
three restores from different snapshots

Matching
Rounds
Scale
Record
Estimate

same host and the same start slot in both conditions
the race test in lockstep rounds, at least 300 per condition
12 snapshot families of three siblings each
binary outcome and failure class for every cell and round
absolute sibling dependence and the sibling excess $\Delta\rho$

It measures environmental dependence, not agent judgments.

Pre-registration, tags prereg-v1 to prereg-v3. Separately prepared restores do not establish independence.

Families and rounds determine the inference

Each design issue needs a declared treatment before data collection.

Design issue	Required treatment
Families reused across comparisons	account for shared families in the test and the interval
Repeated rounds	check temporal dependence between rounds
Host and scheduling	control or randomise the assignment
Restore failure	predeclare replacement and reporting
Excess correlation	keep the baseline correlation explicit

Calculation. Nine measurements, $N_{\text{eff}} = 9/(1 + 8\rho)$.

Strangers, $\rho = 0.10$ $N_{\text{eff}} = 5.0$

Siblings, $\rho = 0.15$ $N_{\text{eff}} = 4.1$

Excess 0.05 with a zero baseline $N_{\text{eff}} = 6.4$

Any amendment is dated and made before data collection.

Pre-registration, tags prereg-v1 to prereg-v3. An independent-block design is a proposed amendment.

Repair resampling returns to the original incident

Descriptive proposal: nine samples of the first repair. Unrun.

Hold fixed	pre-repair input, prompt and tool policy (no shell), hosted model endpoint
Record	out-of-plan edit count, distinct diff count, claimed test evidence
Prediction	at least 5 of 9 samples repeat the out-of-plan edit
Additional validation	freeze the disputed patch and deterministically exercise same-owner adoption

Hosted-model sampling lies outside the guest snapshot. Nine samples describe repeatability. They cannot show an effect induced by the snapshot.

The original in-cell pre-repair commit is unavailable. Its public substitute is the parent of the delivered repair, chosen by commit order. No tree-hash comparison confirms that they match.

The resampling describes repeatability of the repair.

Pre-registration, tags prereg-v1 to prereg-v3. Our incident: first work order, 27 Sep 2026, run records.

The redesigned workflow addresses each observed failure

Proposed workflow. It has not been run.

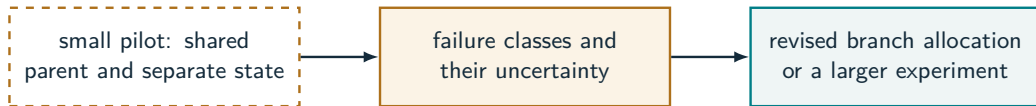
Observed failure	Changed obligation
Out-of-plan repair	scope amendment before acceptance
Missing direct test	explicit behavioural test for the new branch
Reused review feedback	frozen candidate and reserved validation
Misleading receipt	typed execution and actor records
Lost patch	durable export before cleanup
Candidate branching	declared ancestry, budget and evaluation

Speed and search quality need separate measurements at equal total cost: prefix cost, sharing, evidence overlap, evaluation budget and composition.

Forking generates alternatives. The test obligation remains.

Paid tries as part of the result: Stroebel, Kapoor & Narayanan, arXiv:2411.17501 (2024).

A pilot can expose shared failure classes



Published comparison. Nine LLM judges from seven model families were unanimous on 319 of 1,000 MNLI items. On those items, 9.1% of the unanimous decisions were wrong.

Agreement coexisted with error in that setting. The figure is not a fork error rate.

A pilot can expose obvious shared failures. It cannot certify low dependence or rare false acceptance.

Probe a few branches before a broad fan-out.

Kohli, [arXiv:2605.29800](https://arxiv.org/abs/2605.29800) (2026 preprint), §5.2, ChaosNLI-MNLI. Canary requests: Dean & Barroso, The Tail at Scale, CACM 2013.

Two concrete open problems

1 Dependence model

Use recorded ancestry, fixtures and feedback to predict outcome dependence. Validate the calibration on unseen task families.

Useful output. An estimator, a conservative bound or a refusal rule that improves held-out calibration.

Extensions. Warm-state placement, side channels, long human waits and effects on remote services.

We ask for help with these two problems.

2 Authority after restore

Reissue child authority with a generation, an expiry and a budget. Reject stale publication across restore and failure.

Useful output. A checked protocol covering clone, expiry, stale-generation and uncertain-effect cases.

What counts as new evidence

Completed

- MEASURED** two recorded work orders
- BUILT** duplicate-evidence rejection
- BUILT** lineage carried with results
- BUILT** artifact-bound publication gate

Still open

- fork-cost benefit on this workload
- environmental sibling dependence
- calibrated confidence from lineage
- restore-generation authority

 **The runtime can identify reused observations and record shared state.**

Turning those records into calibrated confidence is the research problem.

We request help with a dependence model and with authority after restore.

arXiv:2607.09689

Pre-registered decision rules

Test 1: does restore beat a warm cache?

Step	Design	Decision
Gate 0	a RAM nonce, snapshot, restore 3 children: does it survive?	if not, every result is “restore fan-out”
Fidelity	deterministic tests, restored vs. direct	any output mismatch voids the arm
H1 cost	cached template vs. restore, cold as reference; $N \in \{3, 6, 12\}$; 20 interleaved reps per arm and N ; makespan to first test results, one clock for both arms; no model calls	supports if the 95% CI of the median gain is above 10 s at every N ; rejects if the 98.3% (Bonferroni) CI is below 10 s at any N

Test 2: do siblings fail together?

Step	Design	Decision
H2 coupling	12 snapshot families $\times$ 3 siblings, each paired with 3 strangers restored from other families, same slot, same host; the race test in lockstep rounds, at least 300 per condition	supports if a one-sided sign-flip test gives $p < 0.05$ and $\Delta\rho > 0.05$; rejects if the 90% CI of $\Delta\rho$ is below 0.05; else inconclusive
The loop	repair 1 re-sampled 9 times from its recorded input; the only model calls	descriptive; predicted: ≥ 5 of 9 repeat the out-of-plan edit

Pre-registration in github.com/zozo123/cam-talk-london-26, tags prereg-v1 to prereg-v3.

Research record: biophysics

PhD, University of Houston, with the Rice Center for Theoretical Biological Physics

Eliaz, Nedelec, Morrison, Levine & Cheung. Insights from graph theory on the morphologies of actomyosin networks with multilinkers. *Phys. Rev. E* 102:062420 (2020). First author.

Liman, . . . , **Eliaz**, . . . , Cheung. The role of the Arp2/3 complex in shaping the dynamics and structures of branched actomyosin networks. *PNAS* 117:10825 (2020).

Li, Liman, **Eliaz** & Cheung. Forecasting avalanches in branched actomyosin networks with network science and machine learning. *J. Phys. Chem. B* 125:11591 (2021).

Zegarra, . . . , **Eliaz**, Impact of hydrodynamic interactions on protein folding rates depends on temperature. *Phys. Rev. E* 97:032402 (2018).

Zhang, Nde, **Eliaz**, Jennings, Cieplak & Cheung. CaXML: Chemistry-informed machine learning explains mutual changes between protein conformations and calcium ions in calcium-binding proteins using structural and topological features. *Protein Science* 34:e70023 (2025).

Thesis

Eliaz. Physics of self-assembly in complex matter. PhD thesis, University of Houston (2020); self-published in book form as *Physics of Complex (Biological) Matter* (2023).

Full author lists and positions: CLAIM-FENCE.md. Meeting abstracts (Biophysical Society, APS March Meeting 2018–2021) are not listed as papers.

Research record: genomics

Genome architecture and bioinformatics

Harris, . . . , **Eliaz**, . . . , Lieberman Aiden, Rowley. Chromatin alternates between A and B compartments at kilobase scale for subgenic organization. *Nat. Commun.* 14:3303 (2023).

Kaushal, . . . , **Eliaz**, CTCF loss has limited effects on global genome architecture in *Drosophila* despite critical regulatory functions. *Nat. Commun.* 12:1011 (2021).

Barkai, . . . , **Eliaz**, OffRisk: a docker image for annotating CRISPR off-target sites in the human genome. *Bioinformatics Advances* 3:vbad138 (2023).

Cancer genomics

Saurty-Seerunghen, El-Habr, **Eliaz**, . . . , Junier. A unique malignant cell type per patient tumor encoded in each cancer cell transcriptome. *iScience* 29:115139 (2026).

Gilad, **Eliaz**, A genome-scale CRISPR Cas9 dropout screen identifies synthetically lethal targets in SRC-3 inhibited cancer cells. *Commun. Biol.* 4:399 (2021).

Gilad, **Eliaz**, Reply to: Target expression is a relevant factor in synthetic lethal screens. *Commun. Biol.* 5:836 (2022).

Gilad, **Eliaz**, Drug-induced PD-L1 expression and cell stress response in breast cancer cells can be balanced by drug combination. *Sci. Rep.* 9:15099 (2019).

Full author lists and positions: CLAIM-FENCE.md.

Research record: preprints, systems and talks

Preprints

Eliaz. Evidence-Aware MapReduce for Forkable Compute. arXiv:2607.09689 (2026). Sole author; code github.com/zozo123/boltzmann-mapreduce. Blackburn, Huber, **Eliaz**, . . . , Lieberman Aiden. Cooperation among an anonymous group protected Bitcoin during failures of decentralization. arXiv:2206.02871 (2022). **Eliaz**, Danovich & Gasic. Poolkeh finds the optimal pooling strategy for a population-wide COVID-19 testing. medRxiv (2020). First author. Hitz, . . . , **Eliaz**, . . . , Cherry. The ENCODE uniform analysis pipelines. bioRxiv (2023). The ENCODE Project Consortium (member). The Encyclopedia of DNA Elements. bioRxiv (2026). Chamo & **Eliaz**. Continuation geometry resolves apparent branch splitting in unequal-mass three-body orbits. ai.viXra:2608.0069 (2026); not peer reviewed.

Systems and open source

Airflow Factory, an open-source Apache Airflow software factory: github.com/zozo123/ariflow-swfactory. This talk's slides and notes: github.com/zozo123/cam-talk-london-26. apache/airflow: sandbox toolset with Docker sbx and OpenSandbox backends (merged, 2026); two more backends (open). openclaw/crabbox, nushell, and sandbox PRs to Spark, Kafka Connect and mecatl (2026). The Sandbox Shift: a field manual for running untrusted code (2026).

Talks, October 2026

Cambridge SRG (this talk), 15 Oct.
RustChinaConf, Shenzhen: Disposable Runners, Warm Cargo Factory, 17 Oct.
Zenity AI Agent Security Summit, New York: Your Agent Escaped Without Escaping the Sandbox, 21 Oct.

References: systems

Fork, snapshot, isolation

Barham et al. Xen and the art of virtualization. SOSP 2003.
Vrable et al. Scalability, fidelity, and containment in the Potemkin virtual honeyfarm. SOSP 2005.
Lagar-Cavilla et al. SnowFlock. EuroSys 2009.
Madhavapeddy et al. Jitsu. NSDI 2015.
Manco et al. My VM is lighter (and safer) than your container. SOSP 2017.
Baumann et al. A fork() in the road. HotOS 2019.
Agache et al. Firecracker. NSDI 2020.
Du et al. Catalyzer. ASPLOS 2020.
Brooker et al. Restoring uniqueness in microVM snapshots. arXiv:2102.12892, 2021.
Ustiugov et al. Benchmarking, analysis, and optimization of serverless function snapshots (REAP). ASPLOS 2021.
Lupu et al. Nephele: cloning unikernel-based VMs. EuroSys 2023.
Clark et al. Live migration of virtual machines. NSDI 2005.

Leases and fencing

Gray & Cheriton. Leases. SOSP 1989.
Burrows. The Chubby lock service. OSDI 2006.
Kleppmann. How to do distributed locking. 2016.

Agent sandboxes, 2026

Dong et al. DeltaBox. arXiv:2605.22781. Yu et al. Shepherd. arXiv:2605.10913. Wu et al. Crab. arXiv:2604.28138.
Guo et al. Planarian: managing agent state with statepoints. arXiv:2609.35366. Zheng et al. arXiv:2608.22928.
Kimi Team (Moonshot AI). Kimi K3: Open Frontier Intelligence. Tech. report, 2026, §4.1.2, §4.2.4, §4.2.6, §5.3.2. DeepSeek-AI. DeepSeek-V4 (DSeC). arXiv:2606.19348.
Wei et al. No provisioned concurrency: fast RDMA-codesigned remote fork (MITOSIS). OSDI 2023. Ao et al. FaaSnap. EuroSys 2022.

Effects, authority, provenance

Nightingale, Chen & Flinn. Speculative execution in a distributed file system. SOSP 2005.
Nightingale et al. Rethink the sync. OSDI 2006.
Cully et al. Remus. NSDI 2008.
Watson et al. Capsicum. USENIX Security 2010.
Watson et al. CHERI. IEEE S&P 2015.
Birgisson et al. Macaroons. NDSS 2014.
Muniswamy-Reddy et al. Provenance-aware storage systems. USENIX ATC 2006.
Pasquier et al. Practical whole-system provenance capture. SoCC 2017.
Torres-Arias et al. in-toto. USENIX Security 2019.
O'Callahan et al. Engineering record and replay for deployability. USENIX ATC 2017.

References: evidence and evaluation

Dependent evidence

Dorfman. The detection of defective members of large populations. *Ann. Math. Stat.* 14(4):436–440, 1943.

Cochran. The combination of estimates from different experiments. *Biometrics* 1954.

Kish. *Survey Sampling*. Wiley, 1965.

Amdahl. Validity of the single processor approach to achieving large scale computing capabilities. *AFIPS SJCC* 1967.

Liang & Zeger. Longitudinal data analysis using generalized linear models. *Biometrika* 1986.

Hedges, Tipton & Johnson. Robust variance estimation in meta-regression with dependent effect size estimates. *Research Synthesis Methods* 2010.

Green, Karvounarakis & Tannen. Provenance semirings. *PODS* 2007.

Cameron, Gelbach & Miller. Bootstrap-based improvements for inference with clustered errors. *REStat* 2008.

Cameron, Gelbach & Miller. Robust inference with multiway clustering. *JBES* 2011.

Glasserman & Yao. Some guidelines and guarantees for common random numbers. *Management Science* 1992.

Stockmayer. Theory of molecular size distribution and gel formation in branched-chain polymers. *J. Chem. Phys.* 11:45, 1943.

Shared errors across models

Kim, Garg, Peng & Garg. Correlated errors in large language models. *ICML 2025*, PMLR 267. [arXiv:2506.07962](#).

Goel et al. Great models think alike and this undermines AI oversight. *ICML 2025*. [arXiv:2502.04313](#).

Evaluation under search

Dean & Barroso. The tail at scale. *CACM* 2013 (hedged requests; also fan-out, good enough, canary requests, mutations).

Dean. Designs, lessons and advice from building large distributed systems (“Numbers everyone should know”). *LADIS keynote*, 2009.

Brown et al. Large Language Monkeys: scaling inference compute with repeated sampling. [arXiv:2407.21787](#), 2024.

Von Arx, Chan & Barnes (METR). Recent frontier models are reward hacking. 5 June 2025.

Zhong, Raghunathan & Carlini. ImpossibleBench. [arXiv:2510.20270](#), 2025.

Dwork et al. The reusable holdout. *Science* 2015.

Stroebl, Kapoor & Narayanan. The limits of inference scaling through resampling (v1: Inference scaling fLaws). [arXiv:2411.17501](#), 2024.

Russo & Zou. Controlling bias in adaptive data analysis using information theory. *AISTATS* 2016.

Shared errors across models: 2026 preprints

Jo, Garg & Raghavan. The subjectivity of monoculture. [arXiv:2602.24086](#), preprint.

Kohli. Nine judges, two effective votes: correlated errors undermine LLM evaluation panels. [arXiv:2605.29800](#), preprint.

Chen. When does combining language models help? A co-failure ceiling on routing, voting, and mixture-of-agents across 67 frontier models. [arXiv:2606.27288](#), preprint.

Bone, Stephany & del Rio-Chanona. Monocultural biases: correlated biases in large language models lead to unequal systemic exclusion rates in hiring. [arXiv:2609.22169](#), preprint.

References: own work and documentation

Speaker and co-authors

Y. Eliaz. Evidence-Aware MapReduce for Forkable Compute. arXiv:2607.09689, 2026.

github.com/zozo123/ariflow-swfactory, runs SELFHOST-2/3, 27 Sep 2026.

Blackburn, Huber, Eliaz et al. Cooperation among an anonymous group protected Bitcoin during failures of decentralization. arXiv:2206.02871, 2022.

Saurty-Seerunghen, El-Habr, Eliaz et al. A unique malignant cell type per patient tumor encoded in each cancer cell transcriptome. iScience 29:115139, 2026.

Hitz, . . . , Eliaz, . . . , Cherry. The ENCODE uniform analysis pipelines. bioRxiv 2023, doi:10.1101/2023.04.04.535623.

Gilad, Eliaz et al. A genome-scale CRISPR Cas9 dropout screen identifies synthetically lethal targets in SRC-3 inhibited cancer cells. Commun. Biol. 4:399, 2021.

Barkai, . . . , Eliaz, OffRisk. Bioinformatics Advances 3:vbad138, 2023.

Eliaz, Danovich & Gasic. Poolkeh. medRxiv 2020.

Speaker's physics work (slides 4 and 44)

Eliaz et al. PRE 102:062420, 2020; Liman et al. PNAS 117:10825, 2020; Li et al. JPCB 125:11591, 2021.

Documentation

OBuilder (github.com/ocurrent/obuilder); day10 (tunbury.org, 16 Feb 2026); Docker Sandboxes security; Firecracker snapshotting, snapshot-support and prod-host-setup docs; CRIU TCP_REPAIR; gVisor checkpoint/restore.

RL and exploration

Ecoffet et al. Go-Explore. Nature 2021.